Started on	Sunday, 12 February 2023, 8:53 PM
State	Finished
Completed on	Sunday, 12 February 2023, 8:53 PM
Time taken	11 secs
Marks	0/81
Grade	**0** out of 100

Question 1

Not answered

Marked out of 6

Screening test for depression

Terminally ill patients in a palliative care service are screened for depression with the question 'Do you think you are depressed?' This is then compared to a structured diagnostic interview (DSM) used to diagnose depression (gold standard). Please see the following table and answer each question underneath using the choices below:

	Gold standard test	
	Depressed	Not depressed
Yes to question	11	9
No to question	14	40
	25	49

What was the prevalence of depression in the sample?	Choose...
What was the sensitivity of the screening?	Choose...
What was the specificity of the screening?	Choose...
What proportion of those who screened positive was depressed?	Choose...
What proportion of those who screened negative was not depressed?	Choose...
The proportion of the sample that was correctly classified using the screening test.	Choose...

Your answer is incorrect.

Explanation:

You may want to add one last column to this table to show the total numbers across the rows.

Prevalence can be deduced using a 2X2 table if you compute a row with total numbers in each group. This row is already given (last row) in this table. Prevalence = 25/(25+49) = 25/74.

Sensitivity and specificity can be calculated using standard formulae: refer to the study notes. Sensitivity = 11/25 = 44% while specificity = 40/49 which is slightly higher than 80%

Qs 4 and 5 express the concepts of positive predictive value and negative predictive value.

Q6 expresses accuracy - a formula is given in your study notes. Add the two sets of true observations - true positive and true negative (11+40 = 51), then divide this by total sample size (74). This 51/74 = 66% gives the accuracy.

The correct answer is: What was the prevalence of depression in the sample? → 33%, What was the sensitivity of the screening? → 44%, What was the specificity of the screening? → >80%, What proportion of those who screened positive was depressed? → 55%, What proportion of those who screened negative was not depressed? → 77%, The proportion of the sample that was correctly classified using the screening test. → 66%

Question 2

Not answered

Marked out of 4

Economic analysis of clozapine

An open-label, randomized controlled trial compared 10 patients on clozapine with 10 patients on physicians'-choice medications. 6 out of 10 patients on clozapine achieve remission while 5 out of 10 patients on other antipsychotics achieve remission. For each patient, clozapine costs 3000£ per course while other antipsychotics cost 1800£ per course. On the whole clozapine-treated group achieved 200 symptom-free months while the other group achieved 150 symptom-free months. Choose one of the above options for each of the following measures:

The cost per patient remitted on clozapine	Choose...
The cost per patient remitted on other antipsychotics	Choose...
The incremental cost of clozapine to achieve one additional remission	Choose...
The incremental cost of clozapine per symptom-free month gained	Choose...

Your answer is incorrect.

Explanation:
1) For 10 patients taking clozapine 3000*10 = 30,000£ are spent i.e. for 6 remissions 30,000 therefore for one remission 30,000/6 = 5000£.
2) For 10 patients taking other antipsychotics 1800*10 = 18,000£ are spent i.e. for 5 remissions 18,000 therefore for one remission 18,000/5 = 3600£.
3) For one additional patient to remit we need to treat 10 patients with clozapine compared to 10 patients with other antipsychotics (i.e. 6 - 5 = 1). The incremental cost is then 30,000 - 18000 = 12,000£.
4) In total 50 months (200-150) are gained by spending 12,000 additional ££. 12,000/50 = 240£ needed for each additional month of remission.

The correct answer is: The cost per patient remitted on clozapine → 5000£, The cost per patient remitted on other antipsychotics → 3600£, The incremental cost of clozapine to achieve one additional remission → 12000£, The incremental cost of clozapine per symptom-free month gained → 240£

Question 3

Not answered

Marked out of 7

Appraisal of xanomeline trial

A randomised controlled trial comparing the efficacy of xanomeline (cholinergic M1 selective agonist) against chlorpromazine (comparative control) was conducted last year. This trial also reported the safety and side effect profile of the two drugs. 500 patients were in each arm receiving either xanomeline or chlorpromazine. In the xanomeline group, 250 patients showed clinical response. In the chlorpromazine group, 200 patients showed clinical response. 100 patients on xanomeline reported extrapyramidal side effects while 50 patients on chlorpromazine group reported EPSEs. Calculate the following and chose the correct option from the below list.

EER for clinical response	Choose...
CER for clinical response	Choose...
Absolute benefit increase for clinical response	Choose...
Number needed to treat for clinical response	Choose...
EER for EPSEs	Choose...
CER for EPSEs	Choose...
Number needed to harm for EPSEs	Choose...

Your answer is incorrect.

Explanation: We need to construct two 2X2 tables for this question.
Table for therapeutic effect:

	Response +ve	No response	
Xanomeline	250	250	500

Chlorpromazine	200	300	500
	450	550	1000

Table for side effects:

	EPSE +ve	No EPSE	
Xanomeline	100	400	500
Chlorpromazine	50	450	500
	150	850	1000

For therapeutic effect:
EER = 250/500 = 50%; CER = 200/500 = 40%
ABI = EER - CER = 10%; 1/ABI = NNT = 100/10 = 10

For side effect:
EER = 100/500 = 20% ; CER = 50/500 = 10%
ARR = EER - CER = 10% ; 1/ABI = NNT = 100/10 = 10

The correct answer is: EER for clinical response → 50%, CER for clinical response → 40%, Absolute benefit increase for clinical response → 10%, Number needed to treat for clinical response → 10, EER for EPSEs → 20%, CER for EPSEs → 10%, Number needed to harm for EPSEs → 10

Question 4

Not answered

Marked out of 5

Fluoxetine for BDD

In a multicentre, double-blind RCT with well-concealed allocation, fluoxetine 20mg was compared to placebo for 200 patients with body dysmorphic disorder. Outcomes (improved vs. not improved in BDD scale) were measured after three months and six months of treatment. 100 patients received fluoxetine while 100 patients received placebo. At six months, 75 patients on fluoxetine improved compared to 25 patients receiving placebo. For ethical reasons further follow-up was avoided, and the placebo group received fluoxetine afterwards. Calculate the following measures using the below study data:

The chance that a patient receiving placebo treatment will improve at six months.	Choose...
The odds of getting better while taking fluoxetine compared to placebo for six months.	Choose...
The absolute benefit increase due to fluoxetine treatment.	Choose...
The number of patients with body dysmorphic disorder that needs treatment to achieve one case of improvement.	Choose...
The rate of improvement when taking fluoxetine.	Choose...

Your answer is incorrect.

Explanation:

Step1: Draw a 2X2 table

	Improved at 6 months	Not improved at 6 months	
Fluoxetine	75	?25	100
Placebo	25	?75	100
	100	100	200

Step 2:
Calculate event rates: Here 'event' is 'getting improved.'
CER: 25/100 (chance that a subject on placebo will improve)
EER: 75/100 (= 3/4 - the rate of improvement when taking fluoxetine)
Step 3: Calculate RR
RR = EER/CER = 75/25 = 3
Step 4: Calculate OR
OR = cross product = 75X75 / 25X25 = 9 (The odds of getting better while taking fluoxetine compared to placebo for 6 months)
Step 5: ARR or BI = CER-EER = 50/100 = 50% (The absolute benefit increase due to fluoxetine treatment)
Step 6: NNT = 1/ARR = 100/50 = 2 (The number of patients with body dysmorphic disorder that needs treatment to achieve one case of improvement)

The correct answer is: The chance that a patient receiving placebo treatment will improve at six months. → 25%, The odds of getting better while taking fluoxetine compared to placebo for six months. → 9, The absolute benefit increase due to fluoxetine treatment. → 50%, The number of patients with body dysmorphic disorder that needs treatment to achieve one case of improvement. → 2, The rate of improvement when taking fluoxetine. → 3/4

Question 5

Not answered

Marked out of 3

Cognitive battery in dementia

A cognitive battery was used to screen for dementia in 2,500 subjects with biopsy-proven dementia on autopsy and in 5,000 age- matched controls that underwent an autopsy on death. The results of the cognitive battery were positive (i.e., dementia was diagnosed) in 1,800 cases and 800 controls, all of whom showed no evidence of dementia at biopsy. Please calculate the following measures:

The sensitivity of the cognitive battery	Choose...
The specificity of the cognitive battery	Choose...
The positive predictive value of the cognitive battery	Choose...

Your answer is incorrect.

Explanation:

Step1: Draw a 2X2 table

	Dementia +	No dementia	
Cognitive battery +	1800 (TP)	800	?2600
Cognitive battery _	?700	?4200 (TN)	?4900
	2500	5000	?7500

Step 2:
Calculate sensitivity and specificity:
Sensitivity: true positive / total diseased = 1800/2500 = 72%
Specificity: true negative / total non-diseased = 4200/5000 = 84%

Step 3: Calculate PPV
PPV = true positive / total positives = 1800/2600 = 69.2%

The correct answer is: The sensitivity of the cognitive battery → 72%, The specificity of the cognitive battery → 84%, The positive predictive value of the cognitive battery → 69.2%

Question 6

Not answered

Marked out of 3

Cannabis and Schizophrenia

The incidence rate of psychosis is 80/100,000 person-years for cannabis smokers and 10/100,000 person-years for non-smokers. The prevalence of cannabis smoking is 10% in the community. Choose one correct answer for each of the following questions.

The relative risk of developing psychosis for cannabis smokers compared with non-smokers	Choose...
The percentage of psychosis that can be attributed to smoking cannabis	Choose...
The excess incidence rate of psychosis that could be averted in the community if cannabis smoking is reduced to 5%	Choose...

Your answer is incorrect.

Explanations:

Step 1: Relative risk = incidence rate in exposed / incidence rate in nonexposed. (We can also express this as RR = EER/CER. Here incidence rates are given which represent risk or event rates.) RR = 80/10 = 8

Step 2: The percentage of psychosis that can be attributed to smoking cannabis is Attributable risk proportion: = AR/ EER (or differences in incidence / incidence in the exposed) = (80-10) X 100 / 80 = 87.5%

Step 3:

If the prevalence of smoking was reduced to 5%, the population attributable risk (PAR) = attributable risk x prevalence of exposure in the population =70/100,000 × 5/100 = 3.5/100,000.

The correct answer is: The relative risk of developing psychosis for cannabis smokers compared with non-smokers → 8, The percentage of psychosis that can be attributed to smoking cannabis → 87.5%, The excess incidence rate of psychosis that could be averted in the community if cannabis smoking is reduced to 5% → 3.5/100,000

Question 7

Not answered

Marked out of 7

First Episode Bipolar Test

Prof. Nomor Riddle is working for the last 25 years to find a diagnostic test for bipolar disorder in patients who have had at least one episode of depression. He collates information from various molecular psychiatric studies and neuroimaging data and designs a battery called First Episode Bipolar Test (FBET). He administers FBET in a blinded fashion to 200 patients with at least one episode of depression followed by use of standard criteria and multimodal information to confirm the diagnosis. He obtains the following results.

	Bipolar	Non bipolar	
FBET +	75	15	90
FBET -	25	85	110
	100	100	200

Choose correct answers for the following questions from the list above:

If a patient is FBET+ how likely is that he has bipolar disorder?	Choose...
What is the prevalence of bipolar disorder among the FBET+ patients?	Choose...
How many patients with a negative FBET will actually not have a diagnosis of bipolar disorder?	Choose...
How many patients without bipolar disorder will be wrongly diagnosed with FBET?	Choose...
How often does the FBET provide a correct result?	Choose...
If a patient has bipolar disorder what is the chance that he will turn FBET positive?	Choose...
If a patient has FBET+ what is the chance that the result is incorrect?	Choose...

Your answer is incorrect.

Explanations: Step 1: Redraw the 2X2 table with all useful indicators:

	Bipolar	Non bipolar	
FBET +	75(TP)	15(FP)	90
FBET -	25(FN)	85(TN)	110
	100	100	200

Step 2: Rephrase questions in a language we can understand!
If a patient is FBET+ how likely is that he has bipolar disorder? - This is Likelihood Ratio of a positive test result. NOTE that this applies to an individual patient, and it is about likelihood.
What is the prevalence of bipolar disorder among the FBET+ patients? - This is positive predictive value
How many patients with a negative FBET will actually not have a diagnosis of bipolar disorder? - This is negative predictive value
How many patients without bipolar disorder will be wrongly diagnosed with FBET? - This is false positivity rate
How often does the FBET provide a correct result? - This is accuracy
If a patient has bipolar disorder what is the chance that he will turn FBET positive? - This is sensitivity
If a patient has FBET+ what is the chance that the result is incorrect? -
Step 3:
Positive predictive value = true positive / total positive = 75/90 (2)
Negative predictive value = true negative/total negative = 85/110 (3)
False positivity rate: false positive / total nondiseased = 15/100 (4)
Accuracy = true positive + true negative / total tested = 160/200 (5)
Sensitivity = true positive / total diseased = 75/100 (6)
Specificity = true negative / total nondiseased = 85/100
Likelihood ratio for a positive test result = sensitivity / (1 - specificity) = 75/15 = 5 (1)
Lastly, 15/90 patients with FBET+ will have incorrect results (7)

The correct answer is: If a patient is FBET+ how likely is that he has bipolar disorder? → 75/15, What is the prevalence of bipolar disorder among the FBET+ patients? → 75/90, How many patients with a negative FBET will actually not have a diagnosis of bipolar disorder? → 85/110, How many patients without bipolar disorder will be wrongly diagnosed with FBET? → 15/100, How often does the FBET provide a correct result? → 160/200, If a patient has bipolar disorder what is the chance that he will turn FBET positive? → 75/100, If a patient has FBET+ what is the chance that the result is incorrect? → 15/90

Question 8

Not answered

Marked out of 5

Domestic abuse and depression

In a study examining the relationship between domestic abuse and depression, women who do and do not suffer abuse are followed up over a five-year period. It is found that 70 of the 500 individuals who undergo domestic abuse develop depression; 150 of 3000 individuals who don't get abused develop depression. Calculate the following:

Incidence of depression among domestic abused	Choose...
Incidence of depression among non-abused individuals	Choose...
Relative risk of depression	Choose...
Risk of depression that can be attributed to domestic abuse	Choose...
Attributable risk percentage	Choose...

Your answer is incorrect.

Explanation:
Step 1:
As usual, draw a 2X2 table

	Depression	No depression	
Abused	70	= 430	500
Not abused	150	= 2850	3000
	= 220	= 3280	3500

Step 2:
Incidence can be determined in cohort studies. Here incidence of depression in exposed = new cases / total exposed = 70/500 (EER) = 0.14
Here incidence of depression in non exposed = new cases / total non exposed = 150/3000 (CER) = 0.05
Relative risk = Incidence exposed / incidence in nonexposed
= 0.14/0.05 = 2.8
Atttributable risk = incidence in exposed - incidence in nonexposed = 0.09
Attributable risk percentage = AR/EER X 100 = 0.09/0.14 X100 = .64

The correct answer is: Incidence of depression among domestic abused → 0.14, Incidence of depression among non-abused individuals → 0.05, Relative risk of depression → 2.8, Risk of depression that can be attributed to domestic abuse → 0.09, Attributable risk percentage → 0.64

Question 9

Not answered

Marked out of 4

Rates and ratios for cannabis use

Answer the following questions using the table shown below. Options in the drop-down menu can be used more than once if required.

	Schizophrenia	No schizophrenia	
Cannabis use	15	120	135
No cannabis use	5	80	85
	20	200	220

If the above table represents results from a case-control study of an inner urban population in Germany, what is the incidence rate among the exposed? Choose...

If the above table represents results from a case-control study of an inner urban population in Germany, what is the odds ratio of exposure? Choose...

If the above table represents results from a cohort study of an inner urban population in Finland, what is the relative risk of schizophrenia in cannabis smokers? Choose...

If the above table represents results from a cross-sectional study of an inner urban population in India, what is the prevalence of schizophrenia? Choose...

Your answer is incorrect.

Explanations:

The incidence rate among exposed = new cases / total exposed = as new cases cannot be determined from a case-control study, we cannot find out this value.
Odds ratio is simply a cross product = 80X15 / 120X5 = 2

Incidence rate among exposed = new cases / total exposed = 15/135 = 1/9. Similarly, incidence rate among nonexposed = new cases / total exposed = 5/85 = 1/17. Relative risk is given by incidence in exposed/ incidence in nonexposed = 17/9.
Prevalence is total number of cases in a population/ total size of population = 20/220

The correct answer is: If the above table represents results from a case-control study of an inner urban population in Germany, what is the incidence rate among the exposed? → Cannot be determined, If the above table represents results from a case-control study of an inner urban population in Germany, what is the odds ratio of exposure? → 2, If the above table represents results from a cohort study of an inner urban population in Finland, what is the relative risk of schizophrenia in cannabis smokers? → 17/9, If the above table represents results from a cross-sectional study of an inner urban population in India, what is the prevalence of schizophrenia? → 20/220

Question 10

Not answered

Marked out of 6

Searching PubMed (Medline)
Lead in: From the options listed below, select the most appropriate term as defined.

What is the default operator term used in PubMed?	Choose...
What would you use to search for citations including a specific phrase?	Choose...
Which operator term is used to eliminate citations from the search?	Choose...
Which operator term is used to retrieve citations which include all the search terms?	Choose...
What would you use to expand a search to include all those terms with a specific text?	Choose...
Which operator term is used to retrieve citations in which each citation includes at least one of the search terms?	Choose...

Your answer is incorrect.

Explanation: The search operator 'AND' is the default operator that retrieves results that include all the search terms. The operator 'NOT' excludes the retrieval of terms from your search.
NOT is useful if you want to eliminate certain citations. The operator 'OR' retrieves results that include at least one of the search terms. Pubmed generally searches for phrases using its MeSH database. If you are using a term that is not in MeSH, then use double quotes to enable phrase search. To search for all terms that begin with a word, enter the word followed by an asterisk (*).

The correct answer is: What is the default operator term used in PubMed? → AND, What would you use to search for citations including a specific phrase? → Double quotes (" "), Which operator term is used to eliminate citations from the search? → NOT, Which operator term is used to retrieve citations which include all the search terms? → AND, What would you use to expand a search to include all those terms with a specific text? → Asterisk (*), Which operator term is used to retrieve citations in which each citation includes at least one of the search terms? → OR

Question 11
Not answered
Marked out of 4

Describing Treatment Benefit
Lead In: Using the descriptions given below, identify the method used to describe treatment benefit:

The number of patients we need to treat with the experimental rather than the control treatment to prevent one of them developing the bad outcome	Choose...
The probability that the observed difference between the treatments is due to chance	Choose...
It gives the risk of an event in one group minus the risk in the other group	Choose...
It gives the risk of an event group divided by the risk in the other group	Choose...

Your answer is incorrect.

Explanation:

Number needed to treat is the number of patients we need to treat with the experimental rather than the control treatment to prevent one of then developing the bad outcome. NNT provides a clear indication of how much therapeutic effort is required to bring one additional good outcome.

P-value is the probability that the observed difference between the treatments is due to chance.

Absolute risk reduction gives the risk of an event in one group minus the risk in the other group. Absolute risk reduction provides a clear indication of clinical significance.

Relative risk gives the risk of an event in one group divided by the risk in the other group. Relative risk provides an indication of how many times better or worse a treatment is, compared to another.

(Ref: Oxford handbook of psychiatry-115)

The correct answer is: The number of patients we need to treat with the experimental rather than the control treatment to prevent one of them developing the bad outcome → Number needed to treat, The probability that the observed difference between the treatments is due to chance → P-value, It gives the risk of an event in one group minus the risk in the other group → Absolute risk reduction, It gives the risk of an event group divided by the risk in the other group → Relative risk

Question **12**

Not answered

Marked out of 5

Effect measures

Lead In: Identify the term for each of the definitions given below:

Risk of experiencing the event in the experimental group relative to the control group	Choose...
The number of patients that need to be treated for one to get the benefit of treatment	Choose...
Difference in risk between the control and experimental group	Choose...
Proportion with the event of interest in the intervention group	Choose...
The proportion by which an intervention reduces the risk of an event	Choose...

Your answer is incorrect.

Explanation:

1) Risk ratio or relative risk is the risk of experiencing the event in the experimental group relative to the control group. RR=EER/CER.

2) Number Needed to Treat (NNT): The number of patients who need to receive the experimental treatment to result in one additional positive event, in comparison with the control treatment. NNT = 1/ARR.

3) Absolute risk reduction is the difference in risk between the control and experimental group. ARR=CER-EER

4) Experimental Event Rate (EER) is the rate of an event (e.g. improvement, side effect, etc.,) in the experimental group. It is calculated as the number of events in the intervention group/total number of subjects in intervention group. Control Event Rate (CER) is the rate of the event in the control group. It is calculated as number of events in the control group/total number of subjects in the control group

5) Relative risk reduction is the proportion of events that would have been avoided in the control group had they been allocated the intervention. RRR = ARR/CER or [1- RR] . In epidemiology, this term is sometimes referred to as attributable risk proportion (ARP).

The correct answer is: Risk of experiencing the event in the experimental group relative to the control group → Risk ratio, The number of patients that need to be treated for one to get the benefit of treatment → Number needed to treat, Difference in risk between the control and experimental group → Absolute risk reduction, Proportion with the event of interest in the intervention group → Experimental event rate, The proportion by which an intervention reduces the risk of an event → Relative risk reduction

Question 13

Not answered

Marked out of 5

Diagnostic test measures

Lead In: Identify the terms used in diagnostic test measures:

The ability of a test to correctly identify true positives among the ill subjects	Choose...
The ability of a test to correctly identify true negatives among the healthy subjects	Choose...
The proportion who are ill among all subjects who are picked up by a test as having the disorder	Choose...
The proportion who are truly well among all subjects who are declared to be well by a test	Choose...
This helps to determine how the result of a diagnostic test influences the probability that the patient has the disorder.	Choose...

Your answer is incorrect.

Explanation:

Sensitivity is the proportion of the 'diseased subjects', who are picked up by the test as having the disorder. In other words, it is the ability of the test to identify those who really have the disorder.

Specificity is the proportion of 'well subjects', who are excluded by the test as not having the disorder. In other words, it is the ability of the test to identify those who really do not have the disorder.

Positive predictive value (PPV) is defined as the proportion of positive test results for which disease is confirmed. In other words, it is the proportion of the test positives who are true positives.

Negative predictive value (NPV) is the proportion of negative test results for which a disease is absent. In other words, it is the proportion of the test negatives who are true negatives.

Likelihood ratio summarises how many times more (or less) likely patients with the disease are to have that particular result than patients without the disease. It is the ratio of the probability of the specific test result in people who do have the disease to the probability in people who do not.

The correct answer is: The ability of a test to correctly identify true positives among the ill subjects → Sensitivity, The ability of a test to correctly identify true negatives among the healthy subjects → Specificity, The proportion who are ill among all subjects who are picked up by a test as having the disorder → Positive predictive value, The proportion who are truly well among all subjects who are declared to be well by a test → Negative predictive value, This helps to determine how the result of a diagnostic test influences the probability that the patient has the disorder. → Likelihood ratio

Question **14**

Not answered

Marked out of 6

Sources of Medical Evidence
Choose the most appropriate resource to access research literature from the below list:

To identify the recommended practice of occupational therapists when accepting referrals for PTSD	Choose...
To identify guidelines on treatment of new onset adolescent depression	Choose...
To identify the evidence reviewing a particular treatment option for cardiac failure in geriatric population	Choose...
To identify the dominant psychoanalyst viewpoint of borderline personality disorder that existed in the 1870s in the United States of America	Choose...
To identify a recent trial comparing the effectiveness of a new drug with an established drug	Choose...
To locate a conference abstract reporting a highly controversial but novel findings on language development in schizophrenia	Choose...

Your answer is incorrect.

The "System for Information on Grey Literature in Europe" (SIGLE) is an European initiative operated by a network of national information or document supply centres active in collecting and promoting grey literature (especially conference abstracts, doctoral thesis etc.).

The CINAHL database is a comprehensive and authoritative source of information for nurses, allied health professionals including OTs, and others interested in health care.

The Cochrane Library contains the Cochrane Database of Systematic Reviews, which is the leading resource for systematic reviews in health care. Systematic reviews are widely accepted as the 'gold standard' resource in terms of evidence-based healthcare information. Cochrane also maintains a trials register.

Psycinfo (maintained by American Psychological Association) has more than 3.4 million records, covering psychology back to its underpinnings in the 17th Century

The correct answer is: To identify the recommended practice of occupational therapists when accepting referrals for PTSD → CINAHL, To identify guidelines on treatment of new onset adolescent depression → NICE, To identify the evidence reviewing a particular treatment option for cardiac failure in geriatric population → Cochrane reviews, To identify the dominant psychoanalyst viewpoint of borderline personality disorder that existed in the 1870s in the United States of America → PsycINFO, To identify a recent trial comparing the effectiveness of a new drug with an established drug → PubMed, To locate a conference abstract reporting a highly controversial but novel findings on language development in schizophrenia → SIGLE

Question **15**

Not answered

Marked out of 3

Reporting guidelines for clinical studies

Choose the most appropriate recommendation to report research studies from the below list:

Reporting of meta-analysis of clinical RCTs	Choose...
Reporting cohort studies	Choose...
Reporting systematic reviews	Choose...

Your answer is incorrect.

The aim of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Statement is to help authors report a wide array of systematic reviews to assess the benefits and harms of a health care intervention.

A group of scientists and editors developed the CONSORT (Consolidated Standards of Reporting Trials) statement to improve the quality of reporting of RCTs.

The Quality of Reporting of Meta-analyses (QUOROM) conference was convened to address standards for improving the quality of reporting of meta-analyses of clinical randomised controlled trials (RCTs).

A group of scientists and editors has developed the STARD (Standards for Reporting of Diagnostic Accuracy) statement to improve the reporting the quality of reporting of studies of diagnostic accuracy.

The STROBE statement (Strengthening the Reporting of Observational Studies in Epidemiology) is a good checklist for preparing a publication of observational studies such as cohort, case-control and cross-sectional studies.

The MOOSE (Meta-analysis of Observational Studies in Epidemiology) statement is intended for the reporting of meta-analyses of observational studies.

Consolidated criteria for reporting qualitative research (COREQ) is a 32-item checklist for interviews and focus groups.

The correct answer is: Reporting of meta-analysis of clinical RCTs → QUOROM, Reporting cohort studies → STROBE, Reporting systematic reviews → PRISMA

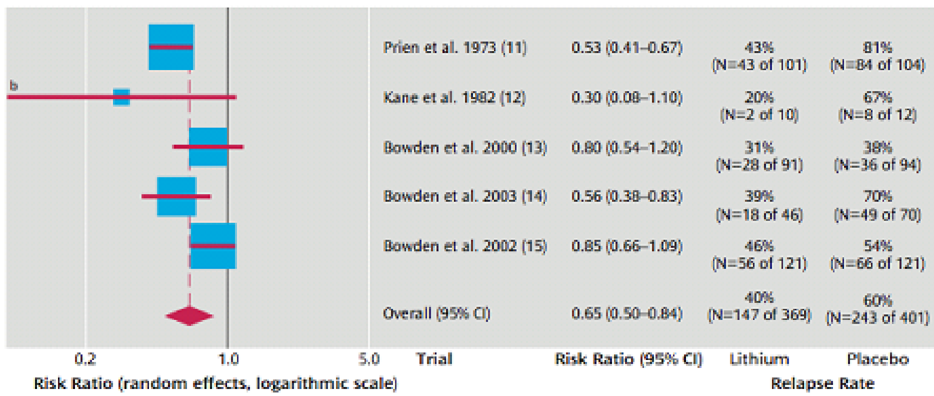
Question **16**

Not answered

Marked out of 4

Relapse in bipolar disorder

FIGURE 1. Randomized, Placebo-Controlled Trials Assessing the Effectiveness of Lithium for the Prevention of Any Relapse in Bipolar Disorder Patients[a]



[a] The area of the blue box represents the weighting given to the trial in the overall pooled estimate and takes into account the number of participants and events and the amount of between-studies variation (heterogeneity).
[b] Lower confidence interval extends beyond graph (0.08).

For each of the following questions choose the correct option from the responses below

The pooled risk of relapse in those taking placebo Choose...

The pooled risk of relapse in those taking lithium Choose...

Pooled risk reduction in lithium group Choose...

Number needed to treat with lithium to prevent relapse Choose...

Your answer is incorrect.

Pooled risk is same as the 'overall risk'. This is obtained by combining all the studies in a meta-analysis. In this plot, we can see that the pooled risk of relapse in lithium group is 40%; in placebo group this is 60%; the different is 20%. The difference is called ARR. 1/ARR = 1/20% = 5 is the NNT.

The correct answer is: The pooled risk of relapse in those taking placebo → 60%, The pooled risk of relapse in those taking lithium → 40%, Pooled risk reduction in lithium group → 20%, Number needed to treat with lithium to prevent relapse → 5

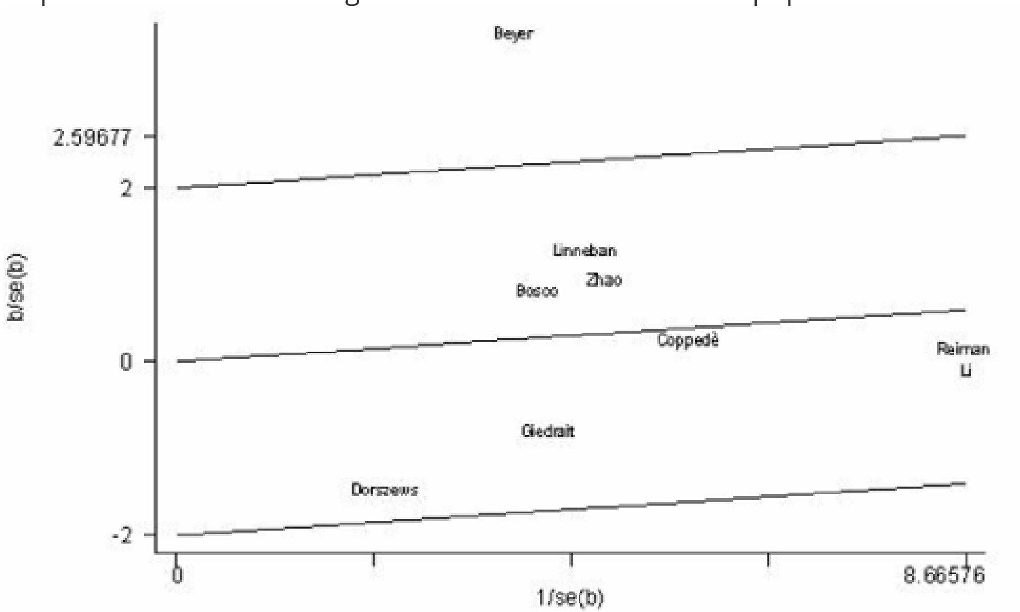
Question 17

Not answered

Marked out of 4

Alzheimer's dementia - Galbraith Plot

A clinical trainee reviews studies reporting associations between MTR A2756G polymorphisms and the risk for Alzheimer's Dementia were reviewed. She identified 9 studies whose result could be statistically synthesized further. She produced the following graph http://www.omicsonline.org/2161-0436/2161-0436-S2-003.php



Choose the study (or studies) from the list that best describes each statement below, from the results shown in the below figure.

The study that can be considered as an outlier	Choose...
The study with a standardised effect size value of -1	Choose...
The study that found comparable levels of Alzheimer's dementia in both AA homozygotes and non-homozygotes.	Choose...

The study that shows the largest magnitude of association between the polymorphism and dementia Choose...

Your answer is incorrect.

The Galbraith plot is designed primarily to assess the extent of heterogeneity between studies in a meta-analysis. The y-axis shows the (log-transformed) effect size divided by its standard error (z-score) and the inverse of the standard error on the x-axis. Each study is represented by a single dot, and a regression line runs centrally through the plot. Parallel to the regression line, at a 2-standard-deviation distance, 2 lines create an interval in which most dots would be expected to fall if the studies were estimating a single fixed parameter. Studies outside the boundaries of 2 SD can be called outliers. Coppede study is close to the 0 line (line of no effect).

See Bax et al., Am. J. Epidemiol. (2009) 169 (2): 249-255. http://aje.oxfordjournals.org/content/169/2/249.long

The correct answer is: The study that can be considered as an outlier → Beyer, The study with a standardised effect size value of -1 → Gleidraitis, The study that found comparable levels of Alzheimer's dementia in both AA homozygotes and non-homozygotes. → Coppede, The study that shows the largest magnitude of association between the polymorphism and dementia → Beyer

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